ORBITAL MEDICAL DIAGNOSTICS

Clinical Laboratory & Aerospace Medicine | Fictional training facility

SYNTHETIC REPORT — NOT A REAL MEDICAL RECORD

PATIENT / CREW ID	AST-001	REPORT DATE	06 Nov 2026
PATIENT	Alex Morgan (fictional)	MISSION DAY	120
SEX / AGE	Male / 35 years	REPORT ID	SIM-2026-1106-001
SPECIMEN PERIOD	05–06 Nov 2026	REPORT TYPE	Comprehensive monitoring

This document is an entirely synthetic software-testing dataset. Results, collection dates and flags are simulated; assay-specific units and clinical reference intervals have not been validated. No clinical interpretation or treatment should be based on this report.

RESULTS BY DEPARTMENT

BONE HEALTH | 63 observations

TEST / OBSERVATION	RESULT	UNIT / SCALE	COLLECTED	DATA FLAG
CTX / β-CTX	0.72	pg/mL ή ng/L	2026-11-05	Review input
NTX	15.93	nmol BCE/mmol creatinine ή assay-specific	2026-11-06	Unvalidated
Pyridinoline (PYD)	6.47	nmol/mmol creatinine	2026-11-06	Unvalidated
Deoxypyridinoline (DPD)	28.63	nmol/mmol creatinine	2026-11-06	Unvalidated
Helical Peptide	19.42	assay-specific	2026-11-06	Unvalidated
Hydroxyproline	18.1	mg/day ή ratio	2026-11-06	Unvalidated
Hydroxylysine	28.27	assay-specific	2026-11-06	Unvalidated
TRAP5b	3.65	U/L ή assay-specific	2026-11-06	Unvalidated
P1NP / PINP	23.99	µg/L	2026-11-06	Unvalidated
PICP	7.47	µg/L	2026-11-05	Unvalidated
BSAP / BAP	8.28	µg/L ή U/L	2026-11-06	Unvalidated
Osteocalcin	17.17	ng/mL	2026-11-06	Unvalidated
Serum Total Calcium	22.45	mg/dL ή mmol/L	2026-11-06	Unvalidated
Ionized Calcium	28.3	mmol/L	2026-11-06	Unvalidated
Urinary Calcium 24h	21.48	mg/day	2026-11-06	Unvalidated
Serum Phosphate	10.19	mg/dL ή mmol/L	2026-11-06	Unvalidated
Urinary Phosphorus	4.73	mg/day	2026-11-06	Unvalidated
PTH	—	pg/mL	2026-11-06	Unvalidated
25(OH) Vitamin D	9.31	ng/mL ή nmol/L	2026-11-05	Unvalidated
1,25(OH)₂ Vitamin D	29.14	pg/mL	2026-11-06	Unvalidated
FGF23	15.56	pg/mL ή RU/mL	2026-11-06	Unvalidated
Sclerostin	15.76	pmol/L ή ng/mL	2026-11-06	Unvalidated

TEST / OBSERVATION	RESULT	UNIT / SCALE	COLLECTED	DATA FLAG
RANKL	4.35	assay-specific	2026-11-06	Unvalidated
OPG	7.05	assay-specific	2026-11-06	Unvalidated
DKK1	26.48	assay-specific	2026-11-06	Unvalidated
Stable Ca isotope ratio	21.33	δ44/42Ca ‰	2026-11-06	Unvalidated
Stable Ca tracer kinetics	3.75	fractional absorption / model	2026-11-06	Unvalidated
Urinary Citrate	26.15	mg/day	2026-11-05	Unvalidated
Urinary Oxalate	14.02	mg/day	2026-11-06	Unvalidated
Urinary Magnesium	12.14	mg/day	2026-11-06	Unvalidated
Urinary Sodium	24.35	mmol/day	2026-11-06	Unvalidated
Urinary Potassium	20.97	mmol/day	2026-11-06	Unvalidated
Urinary Sulfate	13.65	mmol/day	2026-11-06	Unvalidated
Urinary Uric Acid	12.13	mg/day	2026-11-06	Unvalidated
Urinary Creatinine	—	mg/day	2026-11-06	Unvalidated
Urine pH	16.78	pH	2026-11-06	Unvalidated
Total Urine Volume	25.2	mL/day	2026-11-05	Unvalidated
Calcium Oxalate Supersaturation	21.54	dimensionless/index	2026-11-06	Unvalidated
Brushite / Ca Phosphate Supersaturation	3.55	dimensionless/index	2026-11-06	Unvalidated
Uric Acid Supersaturation	1.86	dimensionless/index	2026-11-06	Unvalidated
Sodium Urate Supersaturation	11.38	dimensionless/index	2026-11-06	Unvalidated
Struvite Supersaturation	9.39	dimensionless/index	2026-11-06	Unvalidated
Total Hip aBMD	23.58	g/cm²	2026-11-06	Unvalidated
Femoral Neck aBMD	18.62	g/cm²	2026-11-06	Unvalidated
Lumbar Spine aBMD	5.03	g/cm²	2026-11-06	Unvalidated
Total vBMD	13.22	mg HA/cm³	2026-11-05	Unvalidated
Cortical vBMD	17.72	mg HA/cm³	2026-11-06	Unvalidated
Trabecular vBMD	14.93	mg HA/cm³	2026-11-06	Unvalidated
Trabecular Thickness	8.49	mm	2026-11-06	Unvalidated
Trabecular Number	21	1/mm	2026-11-06	Unvalidated
Trabecular Separation	9.15	mm	2026-11-06	Unvalidated
Cortical Thickness	—	mm	2026-11-06	Unvalidated
Bone Failure Load	29.83	N	2026-11-06	Unvalidated
Estimated Bone Strength	11.99	N ή model-specific	2026-11-06	Unvalidated
Resistance Exercise Adherence	15.56	% sessions	2026-11-05	Unvalidated
Resistance Exercise Load	24.31	N ή device-specific kg-equivalent· όχι ανεξέλ	2026-11-06	Unvalidated
Dietary Calcium Intake	11.2	mg/day	2026-11-06	Unvalidated
Dietary Vitamin D Intake	27.13	IU/day	2026-11-06	Unvalidated

TEST / OBSERVATION	RESULT	UNIT / SCALE	COLLECTED	DATA FLAG
Dietary Protein Intake	17.03	g/day	2026-11-06	Unvalidated
Dietary Sodium Intake	16.72	mg/day	2026-11-06	Unvalidated
Fluid Intake	7.88	mL/day	2026-11-06	Unvalidated
Mission Duration	10.57	days	2026-11-06	Unvalidated
Microgravity Exposure	16.29	hours/days	2026-11-06	Unvalidated

CARDIOVASCULAR HEALTH | 93 observations

TEST / OBSERVATION	RESULT	UNIT / SCALE	COLLECTED	DATA FLAG
Resting heart rate	81	bpm	2026-11-05	Review input
Active heart rate	71.93	bpm	2026-11-06	Unvalidated
Maximum heart rate	103.1	bpm	2026-11-06	Unvalidated
Heart rate recovery 1 min	106.56	bpm	2026-11-06	Unvalidated
HRV RMSSD	24	ms	2026-11-06	Review input
HRV SDNN	—	ms	2026-11-06	Unvalidated
HRV LF/HF	1.7	ratio	2026-11-06	Unvalidated
Systolic blood pressure	138	mmHg	2026-11-06	Review input
Diastolic blood pressure	88	mmHg	2026-11-06	Review input
Mean arterial pressure	29.32	mmHg	2026-11-05	Unvalidated
Pulse pressure	8.18	mmHg	2026-11-06	Unvalidated
Orthostatic systolic change	13.02	mmHg	2026-11-06	Unvalidated
Orthostatic diastolic change	21.42	mmHg	2026-11-06	Unvalidated
Orthostatic heart rate change	92.39	bpm	2026-11-06	Unvalidated
SpO2	96	%	2026-11-06	Review input
Respiratory rate	29.29	breaths/min	2026-11-06	Unvalidated
Body temperature	14.54	°C	2026-11-06	Unvalidated
ECG PR interval	72.15	ms	2026-11-06	Unvalidated
ECG QRS duration	323.63	ms	2026-11-05	Unvalidated
ECG QT interval	87	ms	2026-11-06	Unvalidated
ECG QTc interval	160.58	ms	2026-11-06	Unvalidated
ECG rhythm	7.45	category	2026-11-06	Unvalidated
Premature atrial contractions	—	count/day	2026-11-06	Unvalidated
Premature ventricular contractions	25	count/day	2026-11-06	Unvalidated
Arrhythmia episodes	7.01	count/day	2026-11-06	Unvalidated
ST segment deviation	24.56	mm	2026-11-06	Unvalidated
Troponin I	4	ng/L	2026-11-06	Review input
Troponin T	18.7	ng/L	2026-11-05	Unvalidated

TEST / OBSERVATION	RESULT	UNIT / SCALE	COLLECTED	DATA FLAG
NT-proBNP	48	pg/mL	2026-11-06	Review input
BNP	37.54	pg/mL	2026-11-06	Unvalidated
CK-MB	11.43	ng/mL	2026-11-06	Unvalidated
Total cholesterol	116.44	mg/dL	2026-11-06	Unvalidated
LDL cholesterol	135	mg/dL	2026-11-06	Review input
HDL cholesterol	74.29	mg/dL	2026-11-06	Unvalidated
Triglycerides	165	mg/dL	2026-11-06	Review input
Apolipoprotein B	41.05	mg/dL	2026-11-06	Unvalidated
Lipoprotein(a)	13.98	nmol/L	2026-11-05	Unvalidated
Fasting glucose	108.03	mg/dL	2026-11-06	Unvalidated
HbA1c	67.92	%	2026-11-06	Unvalidated
Fasting insulin	—	µIU/mL	2026-11-06	Unvalidated
HOMA-IR	0.82	ratio	2026-11-06	Unvalidated
High sensitivity CRP	14.36	mg/L	2026-11-06	Unvalidated
D-dimer	7.75	mg/L FEU	2026-11-06	Unvalidated
Fibrinogen	117.15	mg/dL	2026-11-06	Unvalidated
Platelet count	21.4	10^3/µL	2026-11-06	Unvalidated
Hemoglobin	15.58	g/dL	2026-11-05	Unvalidated
Hematocrit	58.46	%	2026-11-06	Unvalidated
Serum sodium	20.69	mmol/L	2026-11-06	Unvalidated
Serum potassium	14.5	mmol/L	2026-11-06	Unvalidated
Serum magnesium	119.22	mg/dL	2026-11-06	Unvalidated
Serum creatinine	123.16	mg/dL	2026-11-06	Unvalidated
eGFR	5.14	mL/min/1.73m²	2026-11-06	Unvalidated
Cardiac output	9.45	L/min	2026-11-06	Unvalidated
Stroke volume	17.21	mL	2026-11-06	Unvalidated
Ejection fraction	59.43	%	2026-11-05	Unvalidated
End diastolic volume	5.84	mL	2026-11-06	Unvalidated
End systolic volume	—	mL	2026-11-06	Unvalidated
Left ventricular mass	21.86	g	2026-11-06	Unvalidated
Carotid intima media thickness	2.52	mm	2026-11-06	Unvalidated
Arterial stiffness PWV	8.16	m/s	2026-11-06	Unvalidated
Augmentation index	60.78	%	2026-11-06	Unvalidated
Baroreflex sensitivity	27.21	ms/mmHg	2026-11-06	Unvalidated
VO2 max	21.62	mL/kg/min	2026-11-06	Unvalidated
Exercise tolerance	64.18	minutes	2026-11-05	Unvalidated
Exercise workload	24.76	W	2026-11-06	Unvalidated

TEST / OBSERVATION	RESULT	UNIT / SCALE	COLLECTED	DATA FLAG
Daily steps	7.46	steps/day	2026-11-06	Unvalidated
Sedentary time	24.72	hours/day	2026-11-06	Unvalidated
Hydration intake	11.38	L/day	2026-11-06	Unvalidated
Urine output	12.28	L/day	2026-11-06	Unvalidated
Fluid shift assessment	3.34	ordinal	2026-11-06	Unvalidated
Lower limb venous ultrasound	9.12	category	2026-11-06	Unvalidated
Peripheral edema score	12.03	0-3	2026-11-06	Unvalidated
Syncope episodes	7	count	2026-11-05	Unvalidated
Chest pain score	—	0-10	2026-11-06	Unvalidated
Palpitations score	5	0-10	2026-11-06	Unvalidated
Dyspnea score	6	0-10	2026-11-06	Unvalidated
Dizziness score	4	0-10	2026-11-06	Unvalidated
Cardiac symptoms present	1	boolean	2026-11-06	Unvalidated
Exercise heart rate variability	244.06	ms	2026-11-06	Unvalidated
Sleep heart rate	78.57	bpm	2026-11-06	Unvalidated
Blood pressure morning	24.59	mmHg	2026-11-06	Unvalidated
Blood pressure evening	10.67	mmHg	2026-11-05	Unvalidated
Cardiac MRI fibrosis	64.23	%	2026-11-06	Unvalidated
Coronary calcium score	12.26	Agatston	2026-11-06	Unvalidated
Endothelial function FMD	80.06	%	2026-11-06	Unvalidated
Vascular resistance	19.7	dyn·s/cm5	2026-11-06	Unvalidated
Blood volume	8.97	L	2026-11-06	Unvalidated
Plasma volume	16.77	L	2026-11-06	Unvalidated
Red cell mass	19.93	L	2026-11-06	Unvalidated
Cardiovascular medication	0	boolean	2026-11-06	Unvalidated
Exercise adherence	—	%	2026-11-05	Unvalidated
Radiation exposure cardiovascular	27.47	mSv	2026-11-06	Unvalidated
Cardiac event history	1	count	2026-11-06	Unvalidated

IMMUNE SYSTEM | 108 observations

TEST / OBSERVATION	RESULT	UNIT / SCALE	COLLECTED	DATA FLAG
WBC	3.6	10^3/µL	2026-11-06	Review input
Absolute neutrophil count	856.65	cells/µL	2026-11-06	Unvalidated
Neutrophils	60.77	%	2026-11-06	Unvalidated
Absolute lymphocyte count	486.82	cells/µL	2026-11-06	Unvalidated
Lymphocytes	93.77	%	2026-11-06	Unvalidated

TEST / OBSERVATION	RESULT	UNIT / SCALE	COLLECTED	DATA FLAG
Monocytes	77.57	%	2026-11-06	Unvalidated
Eosinophils	55.16	%	2026-11-05	Unvalidated
Basophils	78.37	%	2026-11-06	Unvalidated
Absolute monocyte count	982.67	cells/µL	2026-11-06	Unvalidated
Absolute eosinophil count	659.19	cells/µL	2026-11-06	Unvalidated
CD3+ T cells	852.28	cells/µL	2026-11-06	Unvalidated
CD4+ T cells	520	cells/µL	2026-11-06	Review input
CD8+ T cells	340	cells/µL	2026-11-06	Review input
CD4/CD8 ratio	2.03	ratio	2026-11-06	Unvalidated
CD19+ B cells	—	cells/µL	2026-11-06	Unvalidated
CD16+ CD56+ NK cells	325.62	cells/µL	2026-11-05	Unvalidated
Naive CD4 T cells	58.69	%	2026-11-06	Unvalidated
Memory CD4 T cells	84.5	%	2026-11-06	Unvalidated
Naive CD8 T cells	72.13	%	2026-11-06	Unvalidated
Memory CD8 T cells	87.67	%	2026-11-06	Unvalidated
Treg cells	73.33	%	2026-11-06	Unvalidated
Activated CD4 T cells	79.04	%	2026-11-06	Unvalidated
Activated CD8 T cells	78.35	%	2026-11-06	Unvalidated
NK cytotoxic activity	86.1	%	2026-11-06	Unvalidated
T cell proliferation	25.95	index	2026-11-05	Unvalidated
B cell proliferation	29.32	index	2026-11-06	Unvalidated
IgG	980	mg/dL	2026-11-06	Review input
IgA	148.26	mg/dL	2026-11-06	Unvalidated
IgM	25.66	mg/dL	2026-11-06	Unvalidated
IgE	21.96	IU/mL	2026-11-06	Unvalidated
IgG1	97.64	mg/dL	2026-11-06	Unvalidated
IgG2	—	mg/dL	2026-11-06	Unvalidated
IgG3	27.2	mg/dL	2026-11-06	Unvalidated
IgG4	56.71	mg/dL	2026-11-05	Unvalidated
Complement C3	113.18	mg/dL	2026-11-06	Unvalidated
Complement C4	24.83	mg/dL	2026-11-06	Unvalidated
CH50	13.13	U/mL	2026-11-06	Unvalidated
CRP	7.2	mg/L	2026-11-06	Review input
ESR	7.34	mm/h	2026-11-06	Unvalidated
IL-1 beta	50.52	pg/mL	2026-11-06	Unvalidated
IL-2	17.83	pg/mL	2026-11-06	Unvalidated
IL-4	23.28	pg/mL	2026-11-06	Unvalidated

TEST / OBSERVATION	RESULT	UNIT / SCALE	COLLECTED	DATA FLAG
IL-6	5.8	pg/mL	2026-11-05	Review input
IL-8	47.79	pg/mL	2026-11-06	Unvalidated
IL-10	40.68	pg/mL	2026-11-06	Unvalidated
IL-12	65.9	pg/mL	2026-11-06	Unvalidated
IL-17	61.21	pg/mL	2026-11-06	Unvalidated
TNF-alpha	63.73	pg/mL	2026-11-06	Unvalidated
IFN-gamma	—	pg/mL	2026-11-06	Unvalidated
TGF-beta	4.45	ng/mL	2026-11-06	Unvalidated
MCP-1	46.35	pg/mL	2026-11-06	Unvalidated
IP-10	79.44	pg/mL	2026-11-05	Unvalidated
GM-CSF	49.73	pg/mL	2026-11-06	Unvalidated
G-CSF	55.14	pg/mL	2026-11-06	Unvalidated
Neopterin	24.13	nmol/L	2026-11-06	Unvalidated
Procalcitonin	10.75	ng/mL	2026-11-06	Unvalidated
Serum cortisol 08:00	29.62	µg/dL	2026-11-06	Unvalidated
DHEA-S	11.62	µg/dL	2026-11-06	Unvalidated
Cortisol/DHEA ratio	1.51	ratio	2026-11-06	Unvalidated
EBV DNA viral load	29.81	copies/mL	2026-11-06	Unvalidated
CMV DNA viral load	2.47	copies/mL	2026-11-05	Unvalidated
VZV DNA viral load	22.51	copies/mL	2026-11-06	Unvalidated
HSV-1 shedding	0	boolean	2026-11-06	Unvalidated
HSV-2 shedding	0	boolean	2026-11-06	Unvalidated
Salivary EBV DNA	23.65	copies/mL	2026-11-06	Unvalidated
Salivary IgA	—	mg/dL	2026-11-06	Unvalidated
Salivary cortisol	10.09	nmol/L	2026-11-06	Unvalidated
Salivary alpha amylase	22.96	U/mL	2026-11-06	Unvalidated
Microbiome alpha diversity	15.44	Shannon	2026-11-06	Unvalidated
Microbiome beta diversity	21.13	index	2026-11-05	Unvalidated
Gut Firmicutes/Bacteroidetes	2.07	ratio	2026-11-06	Unvalidated
Fecal calprotectin	22.54	µg/g	2026-11-06	Unvalidated
Skin microbiome diversity	23.4	Shannon	2026-11-06	Unvalidated
Oral microbiome diversity	7.39	Shannon	2026-11-06	Unvalidated
Blood culture	23.19	category	2026-11-06	Unvalidated
Urine culture	20.19	category	2026-11-06	Unvalidated
Respiratory pathogen PCR	6.02	category	2026-11-06	Unvalidated
Body temperature	14.9	°C	2026-11-06	Unvalidated
Fever episodes	14.78	count/week	2026-11-05	Unvalidated

TEST / OBSERVATION	RESULT	UNIT / SCALE	COLLECTED	DATA FLAG
Infection symptoms score	5	0-10	2026-11-06	Unvalidated
Upper respiratory symptoms	7	0-10	2026-11-06	Unvalidated
Wound healing score	3	0-10	2026-11-06	Unvalidated
Allergy symptoms score	—	0-10	2026-11-06	Unvalidated
Vaccination antibody titer	3.83	AU/mL	2026-11-06	Unvalidated
Vaccine response	92.4	%	2026-11-06	Unvalidated
Oxidative stress 8-OHdG	42.37	ng/mL	2026-11-06	Unvalidated
Malondialdehyde	2.74	µmol/L	2026-11-06	Unvalidated
Total antioxidant capacity	2.59	mmol/L	2026-11-05	Unvalidated
Glutathione reduced	14.94	µmol/L	2026-11-06	Unvalidated
Glutathione oxidized	12.36	µmol/L	2026-11-06	Unvalidated
GSH/GSSG ratio	1.4	ratio	2026-11-06	Unvalidated
DNA damage gamma-H2AX	13.89	foci/cell	2026-11-06	Unvalidated
Radiation dose	8.96	mSv	2026-11-06	Unvalidated
Sleep duration immune context	6.75	hours	2026-11-06	Unvalidated
Exercise load immune context	11.77	minutes/day	2026-11-06	Unvalidated
Dietary protein immune context	10.97	g/day	2026-11-06	Unvalidated
Vitamin D immune context	36.34	ng/mL	2026-11-05	Unvalidated
Zinc	17.54	µg/dL	2026-11-06	Unvalidated
Selenium	28.88	µg/L	2026-11-06	Unvalidated
Ferritin	—	ng/mL	2026-11-06	Unvalidated
Transferrin saturation	92.43	%	2026-11-06	Unvalidated
Albumin	25.76	g/dL	2026-11-06	Unvalidated
Immune adverse events	4	count	2026-11-06	Unvalidated
Immune medication	1	boolean	2026-11-06	Unvalidated
Immunosuppressant use	1	boolean	2026-11-06	Unvalidated
Antibiotic exposure	21.18	days/month	2026-11-05	Unvalidated
Cytokine panel QC	12.02	category	2026-11-06	Unvalidated
Sample processing delay	75.73	minutes	2026-11-06	Unvalidated

BEHAVIORAL & PSYCHOLOGICAL HEALTH | 100 observations

TEST / OBSERVATION	RESULT	UNIT / SCALE	COLLECTED	DATA FLAG
Sleep duration	5.1	hours	2026-11-06	Review input
Sleep efficiency	76	%	2026-11-06	Review input
Sleep latency	38	minutes	2026-11-06	Review input
Wake after sleep onset	79.76	minutes	2026-11-06	Unvalidated

TEST / OBSERVATION	RESULT	UNIT / SCALE	COLLECTED	DATA FLAG
Number of awakenings	2	count	2026-11-06	Unvalidated
REM sleep duration	31.89	minutes	2026-11-06	Unvalidated
Deep sleep duration	22.48	minutes	2026-11-05	Unvalidated
Light sleep duration	19.07	minutes	2026-11-06	Unvalidated
Sleep midpoint	—	hour	2026-11-06	Unvalidated
Sleep regularity index	19.47	0-100	2026-11-06	Unvalidated
Sleep debt	5.98	hours	2026-11-06	Unvalidated
Circadian phase shift	6.94	hours	2026-11-06	Unvalidated
Melatonin onset	12.17	hour	2026-11-06	Unvalidated
Actigraphy rest activity amplitude	21.06	index	2026-11-06	Unvalidated
Sleep quality	7	0-10	2026-11-06	Unvalidated
Insomnia severity index	4.1	0-28	2026-11-05	Unvalidated
Epworth sleepiness scale	5.94	0-24	2026-11-06	Unvalidated
Karolinska sleepiness scale	24.58	1-9	2026-11-06	Unvalidated
Fatigue score	8	0-10	2026-11-06	Review input
Mental fatigue	5	0-10	2026-11-06	Unvalidated
Physical fatigue	4	0-10	2026-11-06	Unvalidated
Energy level	4	0-10	2026-11-06	Unvalidated
Mood score	3	0-10	2026-11-06	Review input
Positive affect	2	0-10	2026-11-06	Unvalidated
Negative affect	5	0-10	2026-11-05	Unvalidated
Perceived stress	8	0-10	2026-11-06	Review input
Anxiety score	7	0-10	2026-11-06	Review input
Depression screening PHQ-9	27.92	0-27	2026-11-06	Unvalidated
Anxiety screening GAD-7	29.96	0-21	2026-11-06	Unvalidated
Irritability score	3	0-10	2026-11-06	Unvalidated
Loneliness score	2	0-10	2026-11-06	Unvalidated
Social isolation score	2	0-10	2026-11-06	Unvalidated
Team cohesion score	4	0-10	2026-11-06	Unvalidated
Interpersonal conflict events	4	count	2026-11-05	Unvalidated
Social interaction duration	19.39	minutes/day	2026-11-06	Unvalidated
Communication quality	4	0-10	2026-11-06	Unvalidated
Workload NASA TLX	16.13	0-100	2026-11-06	Unvalidated
Cognitive workload	2	0-10	2026-11-06	Unvalidated
Task completion rate	61.74	%	2026-11-06	Unvalidated
Task error rate	95.99	%	2026-11-06	Unvalidated
Attention score	17.28	0-100	2026-11-06	Unvalidated

TEST / OBSERVATION	RESULT	UNIT / SCALE	COLLECTED	DATA FLAG
Reaction time	340	ms	2026-11-06	Review input
Psychomotor vigilance lapses	7	count	2026-11-05	Review input
PVT mean reaction time	250.14	ms	2026-11-06	Unvalidated
PVT fastest 10 percent	43.28	ms	2026-11-06	Unvalidated
PVT slowest 10 percent	236.28	ms	2026-11-06	Unvalidated
Working memory n-back accuracy	87.26	%	2026-11-06	Unvalidated
Working memory response time	71.53	ms	2026-11-06	Unvalidated
Stroop interference	130.49	ms	2026-11-06	Unvalidated
Digit symbol substitution score	4	count	2026-11-06	Unvalidated
Spatial orientation errors	5	count	2026-11-06	Unvalidated
Decision making accuracy	77.1	%	2026-11-05	Unvalidated
Executive function score	17.52	0-100	2026-11-06	Unvalidated
Risk taking score	6	0-10	2026-11-06	Unvalidated
Psychomotor coordination score	13.53	0-100	2026-11-06	Unvalidated
Sustained attention score	20.27	0-100	2026-11-06	Unvalidated
Cognitive flexibility score	27.33	0-100	2026-11-06	Unvalidated
Memory recall score	82.94	%	2026-11-06	Unvalidated
Situational awareness score	15.13	0-100	2026-11-06	Unvalidated
Motivation score	—	0-10	2026-11-06	Unvalidated
Resilience score	8	0-10	2026-11-05	Unvalidated
Coping ability	2	0-10	2026-11-06	Unvalidated
Perceived control	2	0-10	2026-11-06	Unvalidated
Emotional regulation	6	0-10	2026-11-06	Unvalidated
Burnout score	8	0-10	2026-11-06	Unvalidated
Headache severity	5	0-10	2026-11-06	Unvalidated
Dizziness severity	2	0-10	2026-11-06	Unvalidated
Nausea severity	5	0-10	2026-11-06	Unvalidated
Motion sickness score	4	0-10	2026-11-06	Unvalidated
Appetite score	4	0-10	2026-11-05	Unvalidated
Daily caloric intake	22.71	kcal	2026-11-06	Unvalidated
Hydration intake	14.25	L/day	2026-11-06	Unvalidated
Caffeine intake	19.04	mg/day	2026-11-06	Unvalidated
Exercise minutes	29.39	minutes/day	2026-11-06	Unvalidated
Exercise adherence	78.86	%	2026-11-06	Unvalidated
Sedentary time	4.17	hours/day	2026-11-06	Unvalidated
Screen time	—	hours/day	2026-11-06	Unvalidated
Work hours	6.26	hours/day	2026-11-06	Unvalidated

TEST / OBSERVATION	RESULT	UNIT / SCALE	COLLECTED	DATA FLAG
Rest breaks	22.65	count/day	2026-11-05	Unvalidated
Shift duration	5.24	hours	2026-11-06	Unvalidated
Time since last rest day	23.17	days	2026-11-06	Unvalidated
Noise exposure	7.44	dBA	2026-11-06	Unvalidated
Cabin light exposure	5.44	lux	2026-11-06	Unvalidated
Blue light evening exposure	27.05	lux	2026-11-06	Unvalidated
Cabin CO2 exposure	17	ppm	2026-11-06	Unvalidated
Environmental temperature	6.8	°C	2026-11-06	Unvalidated
Privacy satisfaction	8	0-10	2026-11-06	Unvalidated
Mission morale	2	0-10	2026-11-05	Unvalidated
Adaptation to confinement	7	0-10	2026-11-06	Unvalidated
Homesickness	7	0-10	2026-11-06	Unvalidated
Daily journal sentiment	27.8	index	2026-11-06	Unvalidated
Behavioral event count	7	count	2026-11-06	Unvalidated
Safety incident count	0	count	2026-11-06	Unvalidated
Psychological support contacts	—	count	2026-11-06	Unvalidated
Self-reported concerns	5.3	category	2026-11-06	Unvalidated
Medication affecting alertness	1	boolean	2026-11-06	Unvalidated
Sleep aid use	1	boolean	2026-11-05	Unvalidated
Operational readiness	12.81	0-100	2026-11-06	Unvalidated
Crew performance assessment	28.4	0-100	2026-11-06	Unvalidated
Daily check-in completed	0	boolean	2026-11-06	Unvalidated

END OF SYNTHETIC REPORT